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Launchpad

Project Proposal and Task Assignment

Launch & Ledger method - Part 1 (forward)

This proposal defines the scope, delivery plan, milestones and ownership for Accesco Living's dual-engine replenishment intelligence system: a warehouse-facing demand-forecasting and deterministic inventory-planning engine, plus a customer-facing XGBoost Survival AFT time-to-depletion engine. Operational metrics, data coverage and resources not yet confirmed are marked TBC; no model accuracy or business performance is assumed.

1. Need and Traction

Problem or opportunity: Accesco needs two complementary replenishment capabilities. The warehouse-facing engine forecasts SKU demand at each warehouse/store/node using TimesFM 3.0, then applies deterministic inventory rules to recommend when and how much to replenish. The customer-facing engine uses XGBoost Survival AFT to estimate the time until a customer is likely to deplete a purchased product and can support a timely reminder when reliable, consented purchase history becomes available.

Evidence of need

Evidence	Source	Figure / status
SKU identifiers and product catalogue exist	Existing SKU/product catalogue; backend data inventory	Catalogue exists; SKU count and field completeness TBC
Operational data readiness needs validation	PostgreSQL/backend: orders, stock, reservations, purchase orders, receipts and lead times	Table/field availability, history length and granularity TBC
Stockout, overstock and expiry baseline not yet quantified	Stockout/cancellation, inventory, expiry/write-off and service-level reports	Baseline metrics to be measured during initiation

Requested by: CDAO-led data and analytics initiative; formal management/operations sponsor and approval to be confirmed.

Cost of not proceeding: Continued reactive replenishment, avoidable lost sales and cancellations, excess working capital, potential expiry/write-off, and limited visibility into SKU-level demand and inventory risk.

Goals

ID	Goal	Measure	Current	Target	By
G1	Establish data readiness	Critical fields mapped; quality report	Catalogue exists; operational coverage TBC	100% of critical fields documented; gaps and owners agreed	Week 4
G2	Validate warehouse demand forecast and inventory plan	MAE/WAPE/bias on chronological holdout; inventory simulation and Operations review	Not yet measured	TimesFM 3.0 evaluated against simple baselines; inventory policy accepted after simulation; customer AFT gated on data readiness	Week 10-12
G3	Validate replenishment decisions	Stockout / overstock / expiry indicators; recommendation review	Baseline and current process TBC	Pilot recommendations reviewed by Operations; improvement target approved after baseline measurement	Week 16

2. Project at a Glance

Field	Details
Project name	Accesco Dual-Engine Replenishment System (MVP)
Department	Data & Analytics, with Operations and Engineering
Sponsor	Management / Operations sponsor - TBC
Project manager	CDAO / Project Lead - named assignment and contact TBC
Start - end date	TBC; proposed 16-week plan after approval
Budget	TBC; estimate after data, staffing, infrastructure and pilot needs are assessed
Priority	High (proposed; approval required)
ERP and project code	Project tracking/ERP tool TBC; project code TBC

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Project Description and Delivery Roadmap

What the project will do: Specify and phase two distinct engines. The warehouse-facing engine combines TimesFM 3.0 demand forecasts with a deterministic inventory planner using usable stock, reservations, confirmed inbound supply, supplier lead time, safety stock, minimum order quantity, pack size and shelf-life/capacity constraints. The customer-facing engine uses XGBoost Survival AFT to estimate time-to-depletion for a customer-product pair, explicitly handling customers whose depletion/replenishment event has not yet been observed (censoring). Model training and customer alerts are gated on verified data readiness; the first warehouse release remains reviewable by Operations.

Engine Overview - Two Complementary Engines

The engines share governed product, inventory and operational data, but they solve different decisions and require separate training labels, validation and release gates.

Engine / selected model	Purpose and core logic	Required data and release gate
Warehouse-facing replenishment: TimesFM 3.0 + deterministic inventory planning	TimesFM 3.0 forecasts demand per SKU and warehouse/store/node over an operational horizon. Deterministic rules then calculate inventory position, demand during lead time, reorder point, order quantity and priority, applying MOQ, pack-size, expiry, capacity and budget constraints.	Needs usable SKU-node demand context plus stock, reservations, confirmed inbound, supplier lead time and operating rules. Evaluate forecasts on chronological local holdouts and compare with simple benchmarks. Operations reviews pilot recommendations.
Customer-facing time-to-depletion: XGBoost Survival AFT	Estimates time until a customer is likely to deplete a purchased product. The survival approach handles right-censored observations when depletion/replenishment has not been observed by the data cutoff. If direct depletion cannot be observed, a repeat-purchase interval is a proxy and must be labelled as such.	Needs linked customer-product purchases, quantity/pack-size normalization, timestamps, event/proxy definitions and censoring labels. Training and personalized reminders are gated on sufficient observed history, model calibration, consent, privacy review and customer controls.

In scope: (1) Audit SKU/product data and relevant PostgreSQL/backend sources; (2) define warehouse SKU x node x time-series data and the TimesFM 3.0 forecasting interface; (3) specify deterministic inventory-position, reorder-point, order-up-to quantity, MOQ/pack, expiry and capacity rules; (4) define the customer-product time-to-depletion label, censored-observation handling and XGBoost Survival AFT feature set; (5) define event collection for purchases, stock/availability and customer consent; (6) benchmark warehouse forecasts against simple seasonal-naive/moving-average baselines when actual time series exist; (7) validate inventory recommendations using backtests/simulation and a human-approved shadow pilot; (8) document data gates, monitoring, fallback and eventual API integration.

Out of scope for the initial release: autonomous purchase-order placement or stock transfers; production customer notifications before linked purchase history, suitable depletion labels, consent and privacy review are available; and multi-echelon optimization. TimesFM 3.0 results will not be treated as validated for Accesco until evaluated on relevant local data. Synthetic or public proxy data may test code paths but cannot prove Accesco-specific performance.

Completed when: (1) the data-readiness report and dual-engine contracts are reviewed; (2) TimesFM 3.0 is evaluated against agreed simple baselines when suitable demand history exists; (3) the deterministic inventory planner returns traceable, constraint-checked recommendations; (4) the customer XGBoost AFT label/features/censoring specification and data-readiness gate are approved, with training only if labels are sufficient; (5) the Operations shadow pilot, evaluation metrics, fallback, monitoring and handover are documented and accepted.

3. Roadmap

Planning assumption: 16 weeks from an approved start date. Timing will be re-baselined once data access, staffing and pilot sites are confirmed.

Phase	Weeks	Key activities	Output
1. Initiation	1-2	Confirm problem, stakeholders, pilot scope, data access, KPIs and decision rights.	Approved proposal, owners and initial data inventory
2. Planning	3-4	Profile data; define data contract, quality thresholds, evaluation split, architecture and pilot acceptance criteria.	Data-readiness report and detailed plan
3. Execution	5-12	Implement deterministic inventory policy; evaluate TimesFM 3.0 when usable SKU-node demand histories exist; define customer AFT event labels/censoring and capture requirements; run chronological backtests and QA.	Warehouse forecasting + deterministic planning MVP, baseline/backtest report, and customer-engine data contract/readiness gate.

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Phase	Weeks	Key activities	Output
4. Launch	13-15	Run shadow pilot; require staff approval; gather operational feedback and monitor forecast and inventory metrics.	Pilot findings and exception log
5. Closure	16	Review goals, document limitations, hand over runbook and recommend go/no-go for production.	Final report / Ledger and handover

Milestones

ID	Milestone	Phase	Target	Owner
M1	Requirements, KPIs and pilot scope agreed	Initiation	End Week 2 (date TBC)	CDAO / Project Lead + sponsor (TBC)
M2	Data audit, data contract and evaluation plan approved	Planning	End Week 4 (date TBC)	Data/ML lead (TBC)
M3	TimesFM 3.0 benchmark, deterministic inventory planning and customer AFT data contract complete	Execution	End Week 12 (date TBC)	Data/ML lead (TBC)
M4	Shadow pilot reviewed against acceptance criteria	Launch	End Week 15 (date TBC)	Operations owner (TBC)
M5	Handover and go/no-go decision recorded	Closure	Week 16 (date TBC)	Project Lead + sponsor (TBC)

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4. Team

Headcount: Not yet assigned. Sponsor, project manager, data/ML, engineering and Operations team members are all yet to be assigned. Role descriptions below are placeholders only; no individual or team is currently assigned.

Role	Name(s) / status	People	Time on project	Responsible for
Sponsor	Yet to be assigned	TBC	TBC	Priorities, approvals, budget/resource decisions and blocker removal
Project manager	Yet to be assigned	TBC	TBC	Plan, scope, task assignment, risk/issues, reporting and coordination
Data / ML lead	Yet to be assigned	TBC	TBC	Dataset, features, forecasting baselines/candidates, validation and model documentation
Data engineer / backend engineer	Yet to be assigned	TBC	TBC	Data extraction/transformation, pipeline, data controls and eventual API integration
Operations / inventory owner	Yet to be assigned	TBC	TBC	Operational rules, review of recommendations and pilot feedback
External/vendor support	Not assigned / if required later	TBC	TBC	Specialist support only if approved after architecture and budget review
Total	Team not yet assigned	TBC	TBC	Headcount and time commitment to be confirmed before execution

5. Task Assignment in the ERP / Project Tracker

ERP / project tracker: TBC. No project tasks have been assigned yet because the project owner, team members and task owners are not yet confirmed. This task-assignment table is intentionally left empty and will be completed at kickoff after management confirms the project sponsor, project manager, team and tracking system.

ID	Task	Milestone	Assigned to	Assigned by	Start	Due	Priority

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6. Working Meetings

These meetings run the work. Progress reporting and task updates remain recorded in the selected ERP/project tracker.

Meeting	When	Length	Attendees	Purpose
Kick-off	Once, Week 1	60 min	Sponsor, CDAO/PM, Data/ML, Engineering and Operations representatives	Confirm objectives, scope, data access, task owners, tracker and decision rights
Weekly project sync	Weekly; time TBC	30 min	CDAO/PM, Data/ML, Engineering and Operations owner	Progress, data quality, forecast results, blockers, decisions and risks
Weekly planning	Weekly; time TBC	45 min	PM and task owners / team leads	Review tracker, sequence tasks, assign capacity and update due dates
Steering / gate review	At M1-M5; otherwise fortnightly	30 min	Sponsor, CDAO/PM, Operations owner	Approve scope/KPIs, review milestone evidence and make go/no-go decisions

Action items from any meeting are entered in the tracker within 24 hours. Changes to scope, due dates or owners are not approved solely through chat/email.

7. Budget and Risks

Budget

Cost item	Planned amount / approach
People (internal time)	TBC after CDAO, data/ML, engineering and Operations capacity are confirmed
Vendors / contractors	TBC; no vendor assumed at proposal stage
Tools, software and licences	Prefer existing PostgreSQL and approved tools where practical; incremental cloud, compute, BI and licence costs TBC
Pilot / equipment / materials	TBC based on selected pilot stores, operational workflow and integration needs
Contingency	10-15% of the estimated project cost, subject to management approval
Total	TBC - complete after staffing, infrastructure and vendor estimates

Risk register

Risk	Likelihood	Impact	Response	Owner
Insufficient or fragmented history for sales, stock or receipts	High	High	Audit data in Weeks 1-4; agree minimum usable fields; start with transparent baselines and document gaps.	Data/ML lead (TBC)
Recorded sales understate demand during stockouts	High	High	Flag stockout periods; do not treat zero availability as zero demand; use sensitivity checks and operational review.	Data/ML lead (TBC)
Supplier lead times / inbound purchase orders missing or unreliable	Medium-High	High	Validate against receipts; use documented provisional defaults/manual overrides until history is adequate.	Operations owner (TBC)
Expiry and shelf-life constraints absent for perishables	Medium	High	Add expiry/shelf-life controls where fields are reliable; exclude unsupported expiry optimization from claims.	Operations owner (TBC)
Backend and production integration path not finalized	High	Medium-High	Deliver first as a controlled report/export or lightweight pilot; approve API/backend design separately.	Project Lead (TBC)
Forecast drift, promotions or limited pilot adoption	Medium	High	Track WAPE/bias and stockout/overstock outcomes; keep human approval; review overrides and retrain only under a defined process.	Project Lead + Operations (TBC)

Approval items to close before kickoff

Before kickoff: assign the sponsor, project manager and team; confirm source tables and data-access owner; select pilot warehouses/stores and SKU categories; agree forecast horizon, time granularity and inventory service policy; confirm how customer-product depletion will be observed or approximated and what consent/privacy controls apply; approve measurement targets after the baseline is known; confirm budget/staffing and pilot output format (report/export first, API later if approved). Customer-facing model deployment remains gated until adequate linked purchase sequences and reliable, consented labels exist.